El Niño Southern Oscillation (ENSO) Glossary

anomaly
The difference between the value of a variable (for example temperature) at a given location and its long term average at that location.

circulation
The flow or motion of a fluid.

climatology
A quantitative description of climate showing the characteristic values of climate variables over a region. Climate refers to the statistical collection of weather conditions over a specified period of time. Note that the climate taken over different periods of time (30 years, 1000 years) may be different.

convection
Mass motions in a field resulting in transport and mixing of the field. More specifically, it refers to motion associated with a rising current of air. coupled model (or coupled atmosphere-ocean model). In the context of climate modeling this usually refers to a numerical model which simulates both atmospheric and oceanic motions and temperatures and which takes into account the effects of each component on the other.

cumulonimbus
A cloud type that is dense and vertically developed and is associated with rain (particularly of a convective nature).

composite
An average that is done according to a specific criteria. For example, one could composite the rainfall at a station for all years where the temperature was much above average.

jet stream
Strong winds concentrated within a narrow zone in the atmosphere. Often used in reference to the axis of maximum mid-latitude westerlies located in the high troposphere.

monsoons
Seasonal winds. They are caused primarily by the greater annual variation in air temperature over large land surfaces compared to ocean surfaces though other factors like land-relief are important.

stationary waves
Waves (flow patterns with periodicity in time and/or space) that are fixed relative to Earth.

storm track
The path followed by the center of a low (of atmospheric pressure). In many cases, multiple storms follow the same storm track.

teleconnection
A strong statistical relationship between weather in different parts of the globe. For example, there appears to be a teleconnection between the tropics and North America during El Niño.

thermocline
As one descends from the surface of the ocean the temperature remains nearly the same as it was at the surface. Soon, however, one encounters a zone in which temperature starts decreasing rapidly with depth. This zone is called the thermocline. The thermocline is important because it can support large scale waves which play a major role in ENSO. In studying the tropical Pacific Ocean, the depth of 20C water ("the 20C isotherm") is often used as a proxy for the depth of the thermocline. Along the equator, the 20C isotherm is typically located at about 50m depth in the eastern pacific, sloping downwards to about 150 m in the western Pacific.

upwelling
In ocean dynamics, the upward motion of sub-surface water toward the surface of the ocean. This is often a source of cold, nutrient-rich water. Strong upwelling occurs along the equator where easterly winids are present. Upwelling also can occur along coastlines, and is important to fisheries in California and Peru.

*definitions adapted from the "Glossary of Meteorology" 1959.